

Lecture 1

Taylor's polynomial of degree n of f at x_0

$$p_n(x) = f(x_0) + \frac{(x-x_0)}{1!}f'(x_0) + \dots + \frac{(x-x_0)^n}{n!}f^{(n)}(x_0) \quad (1.2)$$

error (1.3)

$$R_{n+1}(x) = \frac{1}{n!} \int (x-t)^n f^{(n+1)}(t) dt = \frac{(x-x_0)^{n+1}}{(n+1)!} f^{(n+1)}(\xi), \quad \xi \text{ between } x \text{ and } x_0$$

Lecture 2

Polynomial Interpolation

Lagrange Interpolation

$$l_i(x) = \prod_{j=0, j \neq i}^n \frac{x-x_j}{x_i-x_j} = \frac{u_i(x)}{u_i(x_i)} = \frac{u_i(x)}{u'(x_i)} \quad (1.4)$$

$$L_n f(x) = \sum_{i=0}^n l_i(x) f(x_i) \quad (1.3)$$

exists $\xi \in (a, b)$ $(R_n f)(x) = \frac{u(x)}{(n+1)!} f^{(n+1)}(\xi) \quad (1.6)$

Barycentric interpolation

$$w_i = \frac{1}{u'(x_i)} = \frac{1}{\prod_{j=0, j \neq i}^n (x_i - x_j)}, \quad i = 0, 1, \dots, n. \quad \text{barycentric weights. (1.2)}$$

$$L_n f(x) = u(x) \sum_{i=0}^n \frac{w_i}{x - x_i} f_i \quad \text{first barycentric formula. (1.3)}$$

$$L_n f(x) = \frac{\sum_{i=0}^n \frac{w_i}{x - x_i} f_i}{\sum_{i=0}^n \frac{w_i}{x - x_i}}, \quad \text{second barycentric formula. (1.4)}$$

Lecture 3

Newton-type methods

Newton's divided difference formula

$$\psi_n(x) = (x-x_0) \dots (x-x_{n-1})(x-x_n) = (x-x_n)\psi_{n-1}(x)$$

$$\psi'_n(x_i) = (x_i-x_n)\psi'_{n-1}(x_i), \quad i = 0, \dots, n-1$$

$$\psi'_n(x_n) = \psi_{n-1}(x_n) \quad (1.5)$$

$$L_n f(x) = \sum_{i=0}^n \frac{\psi_n(x)}{(x-x_i)\psi'_n(x_i)} f_i \quad (1.6)$$

Newton's divided difference form / Newton's interpolation polynomial (1.12)

$$L_n f(x) = f(x_0) + f[x_0, x_1](x-x_0) + \dots + f[x_0, \dots, x_n](x-x_0) \dots (x-x_{n-1})$$

$$R_n f(x) = \frac{(x-x_0) \dots (x-x_n)}{(n+1)!} f^{(n+1)}(\xi), \quad \xi \in (a, b)$$

Newton's forward and backward difference formula $x_i = x_0 + ih, i = 0, 1, \dots$

Newton's forward difference formula. $s = (x-x_0)/h$

$$\binom{s}{k} = \frac{s(s-1) \dots (s-k+1)}{k!}$$

$$L_n f(x) = f_0 + \binom{s}{1} \Delta f_0 + \binom{s}{2} \Delta^2 f_0 + \dots + \binom{s}{n} \Delta^n f_0 \quad (1.16)$$

$$f(x) - L_n f(x) = h^{n+1} \binom{s}{n+1} f^{(n+1)}(\xi_x), \quad (1.17)$$

Newton's backward difference formula. $s = (x-x_n)/h$

$$L_n f(x) = f_n + \binom{s}{1} \nabla f_n + \binom{s+1}{2} \nabla^2 f_n + \dots + \binom{s+n-1}{n} \nabla^n f_n \quad (1.18)$$

$$f(x) - L_n f(x) = h^{n+1} \binom{s+n}{n+1} f^{(n+1)}(\xi_x), \quad (1.19)$$

Derivate

$$(x^n)' = nx^{n-1}$$

$$(\cos x)' = -\sin x$$

$$(\log_a x)' = \frac{1}{x \ln a}$$

$$(a^x)' = a^x \ln a$$

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

$$(fg)' = f'g + fg'$$

Integrale nedefinite

$$\int \ln x dx = x \ln x - x + C$$

$$\int a^x dx = \frac{a^x}{\ln a} + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

$$\int \frac{dx}{x} = \ln|x| + C$$

$$\int f(x)f'(x) dx = \frac{F^2(x)}{2} + C$$

$$\int f(x)g'(x) dx = f(x)g(x) - \int f'(x)g(x) dx$$

$$\int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + C$$

$$\int \frac{f(x)}{F(x)} dx = \ln|F(x)| + C$$

Lecture 5

Birkhoff Interpolation

$$P^{(j)}(x_k) = f^{(j)}(x_k), \quad k = \overline{0, m}, \quad j \in I_k.$$

$$B_n f(x) = \sum_{k=0}^m \sum_{j \in I_k} b_{kj}(x) f^{(j)}(x_k) \quad (1.2)$$

$$\delta_{jp} = \begin{cases} 0, & j \neq p \\ 1, & j = p \end{cases} \quad (1.3)$$

$$b_{kj}^{(p)}(x_\nu) = 0, \quad \nu \neq k, \quad p \in I_\nu,$$

$$b_{kj}^{(p)}(x_k) = \delta_{jp}, \quad p \in I_k, \quad \text{for } j \in I_k \text{ and } \nu, k = 0, 1, \dots, m,$$

[Peano] $\dots$ **Peano kernel.**

$$L f = \int_a^b K_n(t) f^{(n+1)}(t) dt, \quad K_n(t) = \frac{1}{(n+1)!} L[(\cdot - t)_+^{n+1}]$$

n-lygel af minimum

$$L f = \frac{1}{(n+1)!} f^{(n+1)}(\xi) L e_{n+1} \quad (1.13)$$

Lecture 2

Divided Differences

first-order divided difference $f[x_0, x_1] = \frac{f(x_1) - f(x_0)}{x_1 - x_0} \quad (4.1)$

divided difference of order n of f at the distinct nodes $x_0, x_1, \dots, x_n$

$$f[x_0, x_1, \dots, x_n] = \frac{f[x_1, x_2, \dots, x_n] - f[x_0, x_1, \dots, x_{n-1}]}{x_n - x_0} \quad (4.3)$$

$$\begin{array}{ccccccc} x_0 & f[x_0] & \nearrow & f[x_0, x_1] & \nearrow & f[x_0, x_1, x_2] & \nearrow & f[x_0, x_1, x_2, x_3] \\ x_1 & f[x_1] & \nearrow & f[x_1, x_2] & \nearrow & f[x_1, x_2, x_3] & & \\ x_2 & f[x_2] & \nearrow & f[x_2, x_3] & & & & \\ x_3 & f[x_3] & & & & & & \end{array}$$

$$(4.4) \quad f[x_0, x_0, \dots, x_0] = \frac{f^{(n)}(x_0)}{n!}.$$

divided difference of order n at the node x_0 , of multiplicity $n+1$

$$f[x_0, x_1, \dots, x_n] = \sum_{i=0}^n \frac{f(x_i)}{u'(x_i)} = \sum_{i=0}^n \frac{f(x_i)}{u_i(x_i)}, \quad (4.5)$$

$$u(x) = (x-x_0)(x-x_1) \dots (x-x_n) \text{ and } u_i(x) = \frac{u(x)}{x-x_i}.$$

Finite Differences $x_i = x_0 + ih, i = 0, 1, \dots, n, h > 0.$

k th-order forward difference of f with step h , at x_i . (4.12)

$$\Delta^k f(x_i) = \Delta^{k-1} f(x_{i+1}) - \Delta^{k-1} f(x_i) = \Delta^{k-1} f_{i+1} - \Delta^{k-1} f_i$$

$$\begin{array}{ccccccc} x_0 & f_0 & \nearrow & \Delta f_0 & \nearrow & \Delta^2 f_0 & \nearrow & \Delta^3 f_0 \\ x_1 & f_1 & \nearrow & \Delta f_1 & \nearrow & \Delta^2 f_1 & & \\ x_2 & f_2 & \nearrow & \Delta f_2 & & & & \\ x_3 & f_3 & & & & & & \end{array}$$

$$\Delta^n f(x_0) = \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} f_k \quad (4.14)$$

backward difference ∇ (4.13)

$$\nabla^k f_i = \nabla^{k-1} f_i - \nabla^{k-1} f_{i-1},$$

Lecture 3

Aitken-type methods

Neville's method.

$$\begin{array}{ccccccc} x_0 & P_0 & & & & & \\ x_1 & P_1 & P_{01} & & & & \\ x_2 & P_2 & P_{12} & P_{012} & & & \\ x_3 & P_3 & P_{23} & P_{123} & P_{0123} & & \end{array}$$

$$\begin{aligned} P_{i,0} &:= f(x_i), \quad i = \overline{0, n}, \\ P_{i,j} &:= \frac{(x_i - x_{i-j})P_{i,j-1} - (x - x_i)P_{i-1,j-1}}{x_i - x_{i-j} - x + x_i} \\ &= \frac{1}{x_i - x_{i-j}} \left| \begin{array}{cc} x_i - x_{i-j} & P_{i,j-1} \\ x - x_i & P_{i-1,j-1} \end{array} \right|, \quad i \geq j > 0. \end{aligned} \quad (1.22)$$

$$P_{i,j} = \tilde{P}_{i-j, i-j+1, \dots, i-1, i}, \quad j = i, i-1, \dots, 0,$$

Aitken's method (1.23)

$$P_{i,0} := f(x_i), \quad i = \overline{0, n},$$

$$P_{i,j+1} := \frac{1}{x_i - x_j} \left| \begin{array}{cc} x_i - x_j & P_{i,j} \\ x - x_i & P_{j,j} \end{array} \right| = \frac{(x - x_j)P_{i,j} - (x - x_i)P_{j,j}}{x_i - x_j}, \quad i > j \geq 0.$$

Lecture 4

Hermite Interpolation

double nodes $H_n f(x) = \sum_{i=0}^m \left[h_{i0}(x) f(x_i) + h_{i1}(x) f'(x_i) \right] \quad (1.3)$

$$h_{i0}(x) = [1 - 2l'_i(x_i)(x-x_i)] [l_i(x)]^2, \quad (1.4)$$

$$h_{i1}(x) = (x-x_i) [l_i(x)]^2, \quad i = 0, \dots, m.$$

Newton's divided difference form

$$N_n(x) = f(x_0) + f[x_0, x_0](x-x_0) + f[x_0, x_0, x_1](x-x_0)^2 + f[x_0, x_0, x_1, x_1](x-x_0)^2(x-x_1) + \dots + f[x_0, x_0, \dots, x_m, x_m](x-x_0)^2 \dots (x-x_{m-1})^2(x-x_m)$$

$$R_n(x) = f(x) - H_n f(x) = [\psi_m(x)] \frac{2 f^{(n+1)}(\xi_x)}{(n+1)!}, \quad \xi_x \in (a, b). \quad (1.9)$$

$$h_{ij}(x) = \frac{(x-x_i)^j}{j!} \left[\sum_{k=0}^{r_i-j} \frac{(x-x_i)^k}{k!} [u_i(x)]_{x=x_i}^{(k)} \right] u_i(x). \quad (1.13)$$

$$N_n f(x) = f(z_0) + f[z_0, z_1](x-z_0) + \dots + f[z_0, \dots, z_n](x-z_0) \dots (x-z_{n-1}), \quad (1.15)$$

$$= \frac{u(x)}{(n+1)!} f^{(n+1)}(\xi_x), \quad \xi_x \in (a, b).$$

Lecture 6

Spline Interpolation

Linear Splines

$$s_1(f; x) = f_i + f[x_i, x_{i+1}](x-x_i) = f_i + \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i} (x-x_i).$$

$$|f(x) - s_1(f; x)| \leq \frac{|(x-x_i)(x-x_{i+1})|}{2!} \max_{x \in [x_i, x_{i+1}]} |f''(x)| \leq \frac{h_i^2}{8} \max_{x \in [x_i, x_{i+1}]} |f''(x)|,$$

Cubic Splines

$s_3[x_i, x_{i+1}] = p_i(x) \in \mathbb{P}_3, \quad i = 1, 2, \dots, n-1,$	$p_i(x) = f_i + m_i(x-x_i) + \frac{f[x_i, x_{i+1}] - m_i}{h_i^2} (x-x_i)^2 + \frac{m_{i+1} - 2f[x_i, x_{i+1}] + m_i}{h_i^3} (x-x_i)^3.$
$s_3(f; x_i) = f_i, \quad i = 1, 2, \dots, n,$	
$s'_3(f; x_i) = m_i, \quad i = 1, 2, \dots, n.$	

$$h_i m_{i-1} + 2(h_{i-1} + h_i) m_i + h_{i-1} m_{i+1} = b_i, \quad i = 2, \dots, n-1,$$

Piecewise cubic Hermite interpolation

$$b_i = 3 \left(h_i f[x_{i-1}, x_i] + h_{i-1} f[x_i, x_{i+1}] \right).$$

1. Complete (clamped) splines. $m_1 = f'(a), m_n = f'(b).$	Finding cubic splines using the second derivatives	Lecture 6
2. Endpoint second derivative splines. $2m_1 + m_2 = 3f[x_1, x_2] - \frac{1}{2}f''(a)h_1,$ $m_{n-1} + 2m_n = 3f[x_{n-1}, x_n] - \frac{1}{2}f''(b)h_{n-1}.$	$s_3 _{[x_i, x_{i+1}]} = p_i(x) \in \mathbb{P}_3, i = 1, 2, \dots, n-1,$ $s_3(f; x_i) = f_i, i = 1, 2, \dots, n,$ $s_3''(f; x_i) = M_i, i = 1, 2, \dots, n.$	
$s_3''(f; a) = s_3''(f; b) = 0, 2m_1 + m_2 = 3f[x_1, x_2],$	$h_{i-1}M_{i-1} + 2(h_{i-1} + h_i)M_i + h_iM_{i+1} = 6(f[x_i, x_{i+1}] - f[x_{i-1}, x_i]),$ $2M_1 + M_2 = 6(f[x_1, x_2] - f'),$ $M_{n-1} + 2M_n = 6(f'_n - f[x_{n-1}, x_n]),$	
3. Natural cubic splines. $m_{n-1} + 2m_n = 3f[x_{n-1}, x_n].$		
4. “Not-a-knot” (deBoor) splines. $h_2^2 m_1 + (h_2^2 - h_1^2) m_2 - h_1^2 m_3 = \beta_1,$ $h_{n-1}^2 m_{n-2} + (h_{n-1}^2 - h_{n-2}^2) m_{n-1} - h_{n-2}^2 m_n = \beta_2,$ $\beta_1 = 2(h_2^2 f[x_1, x_2] - h_1^2 f[x_2, x_3]),$ $\beta_2 = 2(h_{n-1}^2 f[x_{n-2}, x_{n-1}] - h_{n-2}^2 f[x_{n-1}, x_n]).$	Least Squares Approximation $a \sum_{i=1}^n x_i^4 + b \sum_{i=1}^n x_i^3 + c \sum_{i=1}^n x_i^2 = \sum_{i=1}^n x_i^2 y_i,$ $a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i = \sum_{i=1}^n x_i y_i$ $a \sum_{i=1}^n x_i^3 + b \sum_{i=1}^n x_i^2 + c \sum_{i=1}^n x_i = \sum_{i=1}^n x_i y_i,$ $a \sum_{i=1}^n x_i + nb = \sum_{i=1}^n y_i.$ $a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i + nc = \sum_{i=1}^n y_i,$	

A_i are called coefficients , $L(f) = \sum_{i=1}^m A_i L_i(f) + R(f), f \in X.$ $\ker R = \mathbb{P}_d \iff \begin{cases} R(e_k) = 0, k = 0, 1, \dots, d, \\ R(e_{d+1}) \neq 0, \end{cases}$	Lecture 7
Numerical Differentiation $f^{(k)}(\alpha) = \sum_{i=0}^m A_i f(x_i) + R(f), \int_a^b f(x) dx = \sum_{k=0}^m \sum_{j \in I_k} A_{kj} f^{(j)}(x_k) + R(f),$	<i>forward difference</i> $f'(x) \approx \frac{f(x+h) - f(x)}{h} \equiv D_h f(x),$
<i>numerical differentiation formula.</i>	<i>numerical integration (quadrature) formula.</i> $(RD_h f)(x) = f'(x) - D_h f(x) = -\frac{h}{2} f''(\xi), \xi \in (x, x+h).$
<i>backward difference</i> $f'(x) \approx \frac{f(x) - f(x-h)}{h} \equiv \tilde{D}_h f(x), (R\tilde{D}_h f)(x) = f'(x) - \frac{f(x) - f(x-h)}{h} = \frac{h}{2} f''(\xi), \xi \in (x-h, x).$	
Method of Undetermined Coefficients $D_h^{(2)} f(x) = \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}.$ $(RD_h^{(2)} f)(x) = f''(x) - D_h^{(2)} f(x) \approx -\frac{h^2}{12} f^{(4)}(x).$	

Numerical Integration $\int_a^b f(x) dx = \sum_{k=0}^m A_k f(x_k) + R(f),$	Trapezoidal Rule $\int_a^b f(x) dx = \frac{b-a}{2} (f(a) + f(b)) - \frac{(b-a)^3}{12} f''(\xi), \xi \in (a, b),$
interpolatory quadrature. $d_{\max} = 2m + 1.$	n subintervals $h = \frac{b-a}{n}.$ composite (repeated) $\int_a^b f(x) dx = \frac{h}{2} [f(a) + 2(f_1 + \dots + f_{n-1}) + f(b)] - \frac{h^2(b-a)}{12} f''(\xi), \xi \in (a, b).$
$\int_a^b f(x) dx = (b-a) f\left(\frac{a+b}{2}\right) + R(f),$	Simpson’s Rule $\int_a^b f(x) dx = \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right] - \frac{(b-a)^5}{2880} f^{(4)}(\xi), \xi \in (a, b).$
Rectangle (Midpoint) Rule $R(f) = \frac{f''(\xi)}{2!} \int \left(x - \frac{a+b}{2}\right)^2 dx = \frac{(b-a)^3}{24} f''(\xi), \xi \in (a, b).$	$\left[h = \frac{b-a}{2m}, n = 2m \right], \int_a^b f(x) dx = \frac{h}{3} \left[f(a) + 4 \sum_{i=1}^m f_{2i-1} + 2 \sum_{i=1}^{m-1} f_{2i} + f(b) \right] - \frac{h^4(b-a)}{180} f^{(4)}(\xi), \xi \in (a, b).$
$\int_a^b f(x) dx = h \sum_{i=0}^{n-1} f\left(a + \left(i + \frac{1}{2}\right)h\right) + \frac{h^2(b-a)}{24} f''(\xi), \xi \in (a, b),$	composite (repeated)

Gaussian Elimination Lecture 9 $x_n = \frac{b_n}{u_{nn}},$ backward substitution: $x_i = \frac{1}{u_{ii}} \left(b_i - \sum_{j=i+1}^n u_{ij} x_j \right), i = \overline{n-1, 1}$	$x_1 = \frac{b_1}{l_{11}},$ forward $x_i = \frac{1}{l_{ii}} \left(b_i - \sum_{j=1}^{i-1} l_{ij} x_j \right), i = \overline{2, n}.$	LU Factorization <i>no row interchanges are necessary</i> $A = LU,$ Schur complement $A' = vw^*/a_{11} = L'U'.$ $A = \begin{bmatrix} a_{11} & w^* \\ v & A' \end{bmatrix}$ $Ly = b \quad \begin{bmatrix} 1 & 0 \\ v/a_{11} & I_{n-1} \end{bmatrix} \begin{bmatrix} a_{11} & w^* \\ 0 & A' - vw^*/a_{11} \end{bmatrix}$ $Ux = y.$
partial pivoting (maximal pivoting on columns). $ a_{pk}^{(k)} = \max_{k \leq l \leq n} a_{lk}^{(k)} .$	total (maximal) pivoting. $ a_{pq} = \max\{ a_{ij} , i, j = \overline{k, n}\}$	LUP Factorization $PA = LU.$ $Ly = Pb$ QR Factorization Lecture 10 $Ux = y.$ $A = QR, Q$ orthogonal and R upper triangular $r_{ii} > 0,$ $Rx = Q^T b.$ Cholesky factorization $A = R^T R,$ $\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ & a_{22} & \dots & a_{2n} \\ & & \ddots & \vdots \\ 0 & & & a_{nn} \end{bmatrix}$
$(R_i) \longleftrightarrow (R_p)$ scaled pivoting on columns: $\frac{ a_{pi} }{s_i} = \max_{1 \leq j \leq n} \frac{ a_{ji} }{s_j}$	$s_i = \max_{j=1, n} a_{ij} $ or $s_i = \sum_{j=1}^n a_{ij} , i = \overline{1, n}.$	

Orthogonal polynomials and recurrence coefficients

Name	Notation	Polynomial	Weight fn.	Interval	α_k	β_k
Legendre	l_m	$[(x^2 - 1)^m]^{(m)}$	1	$[-1, 1]$	0	$\beta_0 = 2,$ $\beta_k = (4 - k^{-2})^{-1}, k \geq 1$
Chebyshev 1 st	T_m	$\cos(m \arccos x)$	$(1 - x^2)^{-\frac{1}{2}}$	$[-1, 1]$	0	$\beta_0 = \pi,$ $\beta_1 = \frac{1}{2},$ $\beta_k = \frac{1}{4}, k \geq 2$
Chebyshev 2 nd	Q_m	$\frac{\sin[(m+1) \arccos x]}{\sqrt{1-x^2}}$	$(1 - x^2)^{\frac{1}{2}}$	$[-1, 1]$	0	$\beta_0 = \frac{\pi}{2},$ $\beta_k = \frac{1}{4}, k \geq 1$
Laguerre	L_m^a	$x^{-a} e^x (x^{m+a} e^{-x})^{(m)}$	$x^a e^{-x}, a > -1$	$[0, \infty)$	$2k + a + 1$	$\beta_0 = \Gamma(1 + a),$ $\beta_k = k(k + a), k \geq 1$
Hermite	H_m	$(-1)^m e^{x^2} (e^{-x^2})^{(m)}$	e^{-x^2}	$\mathbb{R}$	0	$\beta_0 = \sqrt{\pi},$ $\beta_k = \frac{k}{2}, k \geq 1$

Common Rootfinding Methods Lecture 11 Bisection Method <i>two-point method,</i> $c_n = \frac{a_n + b_n}{2}$ Secant Method <i>two-point method,</i> $x_{n+1} = x_n - f(x_n) \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}, n = 1, 2, \dots,$ Newton’s Method $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}, n = 0, 1, \dots$ <i>one-step</i>	Newton’s Method for Multiple Roots Lecture 12 $x_{n+1} = x_n - m \frac{f(x_n)}{f'(x_n)}.$ Newton’s Method for Nonlinear Systems $x_{n+1} = x_n - (J_f(x_n))^{-1} f(x_n), n \geq 0,$ $J_{J_f(x_n)} = \begin{bmatrix} \frac{\partial f_1(x)}{\partial x_1} & \frac{\partial f_1(x)}{\partial x_2} \\ \frac{\partial f_2(x)}{\partial x_1} & \frac{\partial f_2(x)}{\partial x_2} \end{bmatrix} J_f(x_n) \delta_{n+1} = -f(x_n),$ $x_{n+1} = x_n + \delta_{n+1}.$
--	---

Adaptive Quadratures Lecture 8 $I \approx \frac{2^p I_{2n} - I_n}{2^p - 1} = I_{2n} + \frac{I_{2n} - I_n}{2^p - 1} \stackrel{\text{not}}{=} R_{2n}$ Richardson’s extrapolation formula. $I - I_{2n} \approx \frac{I_{2n} - I_n}{2^p - 1},$ <i>error estimate</i>	Iterated Quadratures Romberg’s Method $R_{k,j} = \frac{4^{j-1} R_{k,j-1} - R_{k-1,j-1}}{4^{j-1} - 1}, k = \overline{2, n}, j = \overline{2, k}.$
$\begin{matrix} R_{1,1} \\ R_{2,1} & R_{2,2} \\ R_{3,1} & R_{3,2} & R_{3,3} \end{matrix} \quad R_{n-1,n-1} - R_{n,n} < \varepsilon.$ $\begin{matrix} \vdots & \vdots & \vdots & \ddots \\ R_{n,1} & R_{n,2} & R_{n,3} & \dots & R_{n,n} \end{matrix} \quad S_{k,1} = R_{k,2}.$	Gaussian Quadratures $\int_a^b w(x) f(x) dx = \sum_{k=1}^m A_k f(x_k) + R_m(f)$ <i>maximum degree of precision, $d = 2m - 1.$</i> $\mu_j = \int_a^b w(x) x^j dx$

Iterative Methods Lecture 10 Jacobi and Gauss-Seidel Methods $A = D - L - U, x^{(k+1)} = T x^{(k)} + c$ Jacobi iteration, $A = M - N.$ $M = D, N = L + U,$ so $T_J = D^{-1}(L + U), c_J = D^{-1}b.$ $x^{(k+1)} = D^{-1}(L + U)x^{(k)} + D^{-1}b, k \in \mathbb{N},$ Gauss-Seidel iteration, $M = D - L, N = U,$ so $T_{GS} = (D - L)^{-1}U, c_{GS} = (D - L)^{-1}b.$ $x^{(k+1)} = (D - L)^{-1}Ux^{(k)} + (D - L)^{-1}b,$	Acceleration methods; SOR Method $M = \frac{D}{\omega} - L, N = \left(\frac{1-\omega}{\omega} D + U\right),$ so $T_\omega = \left(\frac{D}{\omega} - L\right)^{-1} \left(\frac{1-\omega}{\omega} D + U\right), c_\omega = \left(\frac{D}{\omega} - L\right)^{-1} b.$ $D = \begin{bmatrix} a_{11} & & & \\ & a_{22} & & \\ & & \ddots & \\ 0 & & & a_{nn} \end{bmatrix} -U = \begin{bmatrix} 0 & a_{12} & \dots & a_{1n} \\ & 0 & \dots & a_{2n} \\ & & \ddots & \vdots \\ 0 & & & 0 \end{bmatrix}$ $-L = \begin{bmatrix} 0 & & & 0 \\ a_{21} & 0 & & \\ \vdots & & \ddots & \\ a_{n1} & a_{n2} & \dots & 0 \end{bmatrix}$
---	--

$f: [a, b] \rightarrow \mathbf{R}$ is continuous c in (a, b) g is an integrable function that does not change sign on $[a, b].$ $\int_a^b f(x) g(x) dx = f(c) \int_a^b g(x) dx$	$\text{tr}(A) = \sum_{i=1}^n a_{ii} = \sum_{i=1}^n \lambda_i = \lambda_1 + \lambda_2 + \dots + \lambda_n$ $\det(A) = \prod_{i=1}^n \lambda_i = \lambda_1 \lambda_2 \dots \lambda_n$
[Banach] (3.2) $g \in C[a, b],$ with $g([a, b]) \subseteq [a, b]$ <i>there exists $0 < \lambda < 1$</i> $ g(x) - g(y) \leq \lambda x - y , \forall x, y \in [a, b]$ g has a unique fixed point $\alpha \in [a, b]$	